

Teaching Exam

Subject – Computer Science

Practice Session.



Lecture No -2

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Topics to be Covered

Topic

Software Engineering.



1. Software is defined as _____

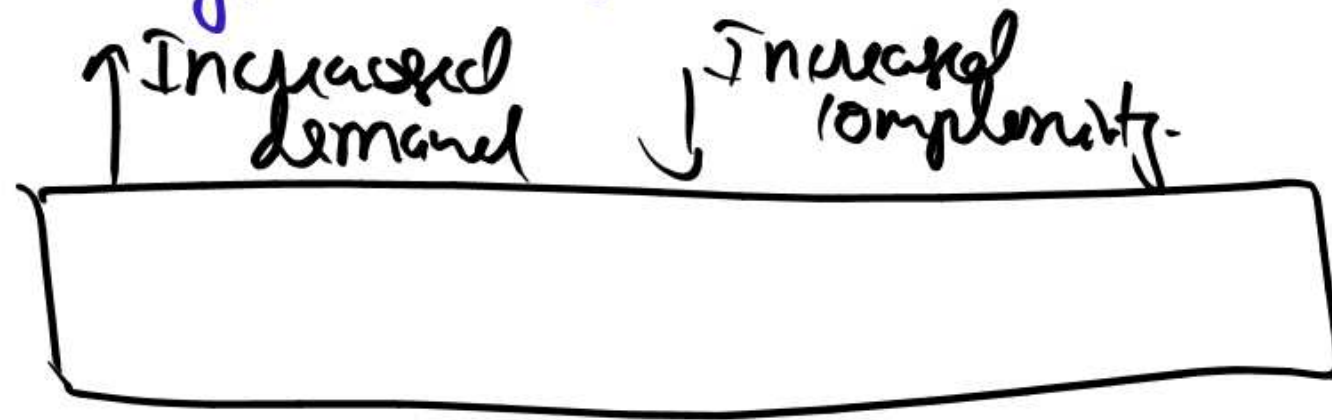
- a) set of programs, documentation & configuration of data
- b) set of programs
- c) documentation and configuration of data
- d) None of the mentioned

Software →

Program Code + Documentation + User manual.

Software Crisis →

- Hardware growth pace. is exponential.
- Software. doesn't match their growth.



Software Engineering

Using engineering principles in the development of Software Product is called Software Engineering.

to ensure $\rightarrow$ correctness, maintainability, low budget, security etc in the process.

Software characteristics →

→ intangible.

→ No wear, Tear.

→ Custom Build

Problems in software development

→ Inadequate information gathering

→ Improper deliver of information by customer.
(Incomplete)

→ Lack of information b/w stakeholders

→ Insufficient Time & Budget

→ Unrealistic deadlines.

→ Resource constraint.

c) Lack of skilled personnel.

Because of rapid changes in technology, personnel becomes outdated in technology talent.

d) Resist to change

Software Quality Attributes

- 1) Correctness = Ability to perform all tasks specified by user.
- 2) Usability = The ease of using the software product.
- 3) Reliability = The performance of the software product should be accurate and should not decrease with time.
- 4) Efficiency = The software product should use hardware resources efficiently.

5) Maintainability → Update,

6) Portability → Software product should be portable among different systems, devices.

7) Scalability → The capability of software to handle increased workload.

8) Security → To stop unauthorized access.



Modularity - construction should be divided in separate modules.

Reusability → The software's component can be reused w/o requiring any no. times.

Testability → The ease by which software can be tested.

Software Process :

1) Feasibility Study :

- 1) Social Issue. //
- 2) Financial benefits //
- 3) Cost and benefit.
- 4) Availability of infrastructure.

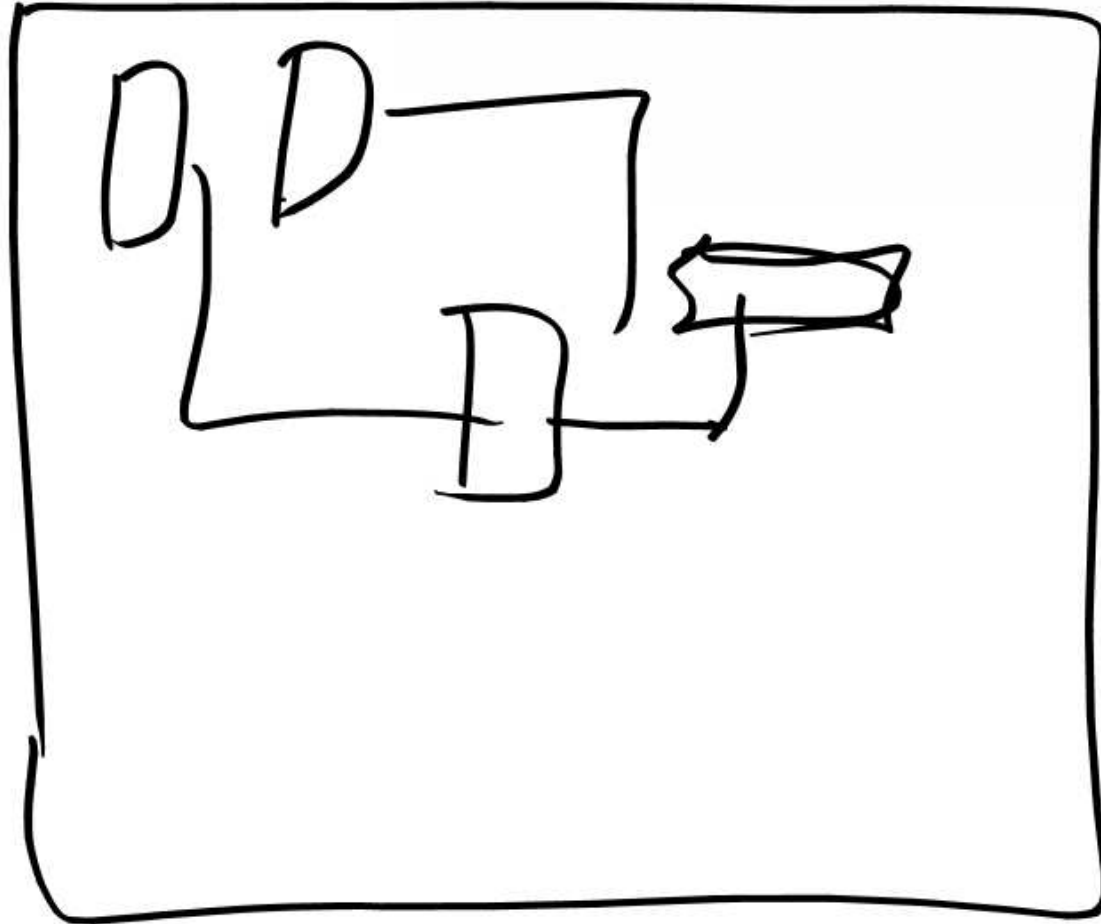
Requirement Analyst's Specification.

SRS is Software Requirement Specifications.

* Interaction with customer, properly to understand their requirement.

* Requirement is collected as carefully otherwise any hearing error can lead to severe impact.

Designing (modelling).



Development (coding)

Implementation of design.
in some coding language.

Testing-s

- Smoke, sanity
- whit box., Black box.
- Manual, Automated

Deployment

Maintenance

SDLC

Waterfall Model. (1970)

Royce.

Small-medium sized software product.

Feasibility Analysis.

Requirement Analysis

Designing

Development

Testing

Deployment.

Feasibility Analysis

Requirement Analysis

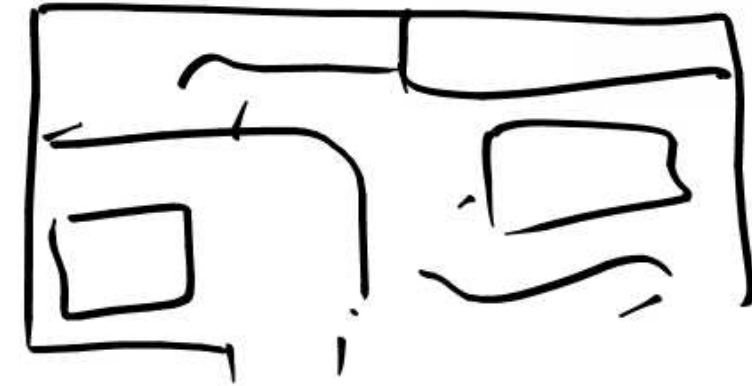
Prototype

Modeling

36k → Area.

20h → 12x12.

1mb = 15x18



2. What is Software Engineering?

- a) Designing a software
- b) Testing a software
- c) Application of engineering principles to the design a software
- d) None of the above

3. Who is the father of Software Engineering?

- a) Margaret Hamilton
- b) Watts S. Humphrey
- c) Alan Turing
- d) Boris Beizer

4. What are the features of Software Code?

- a) Simplicity
- b) Accessibility
- c) Modularity
- d) All of the above

5. _____ is a software development activity that is not a part of software processes.

- a) Validation
- b) Specification
- c) Development
- d) Dependence

6. Define Agile scrum methodology.

- a) project management that emphasizes incremental progress
- b) project management that emphasizes decremental progress
- c) project management that emphasizes neutral progress
- d) project management that emphasizes no progress

7. CASE stands for

- a) Computer-Aided Software Engineering
- b) Control Aided Science and Engineering
- c) Cost Aided System Experiments
- d) None of the mentioned

8. _____ is defined as the process of generating analysis and designing documents?

- a) Re-engineering
- b) Reverse engineering
- c) Software re-engineering
- d) Science and engineering

9. The activity that distributes estimated effort across the planned project duration by allocating the effort to specific software developing tasks is

-
- a) Project scheduling
 - b) Detailed schedule
 - c) Macroscopic schedule
 - d) None of the mentioned

10. What is a Functional Requirement?

- a) specifies the tasks the program must complete
- b) specifies the tasks the program should not complete
- c) specifies the tasks the program must not work
- d) All of the mentioned

11. Why do bugs and failures occur in software?

- a) Because of Developers
- b) Because of companies
- c) Because of both companies and Developers
- d) None of the mentioned



13. The Cleanroom philosophy was proposed by _____

- a) Linger
- b) Mills
- c) Dyer
- d) All of the Mentioned

14. What does SDLC stands for?

- a) System Design Life Cycle
- b) Software Design Life Cycle
- c) Software Development Life Cycle
- d) System Development Life cycle

16. _____ is not among the eight principles followed by the Software Code of Ethics and Professional Practice.

- a) PRODUCT
- b) ENVIRONMENT
- c) PUBLIC
- d) PROFESSION

17. Which of the following are CASE tools?

- a) Central Repository
- b) Integrated Case Tools
- c) Upper Case Tools
- d) All of the mentioned

18. _____ suits the Manifesto for Agile Software Development.

- a) Customer collaboration
- b) Individuals and interactions
- c) Working software
- d) All of the mentioned

19. Software patch is defined as _____

- a) Daily or routine Fix
- b) Required or Critical Fix
- c) Emergency Fix
- d) None of the mentioned

20. _____ software development team has no permanent leader.

- a) Controlled Centralized (CC)
- b) Controlled decentralized (CD)
- c) Democratic decentralized (DD)
- d) None of the mentioned

22. _____ is not a fundamental activity for software processes in software development.

- a) Evolution
- b) Design and implementation
- c) Validation
- d) Verification

23. What are agile manifesto principles?

- a) Customer satisfaction
- b) Face-to-face communication within a development team
- c) Changes in requirements are welcome
- d) All of the mentioned



2 mins Summary



TEACHING
WALLAH

Topic

Spinal.

Agile.

SRS.

Testing

CASE

Cocomo

THANK YOU